

Hadi ZareZadeh

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SUMMARY

Computer Science graduate with strong foundations in algorithms, optimization, and software engineering, now transitioning toward Machine Learning and AI research. Passionate about deep learning, computer vision, NLP, and reinforcement learning, with a growing portfolio of Python-based projects involving automation, data processing, and algorithmic problem-solving. Motivated by long-term research-driven challenges and seeking opportunities to develop advanced AI systems while contributing to innovative and impactful projects.

INTERESTS

Deep Learning, Computer Vision, NLP & Transformer Models, Reinforcement Learning, Unstructured Data Processing, Representation Learning, Optimization Algorithms, Robotics & Autonomous Systems, Algorithmic Trading, Applied AI Research

EDUCATION

Kashmar Higher Education Institute, Kashmar, Iran B.Sc. of Computer Science	Fall 2016 – Summer 2020
Dr. Shariati School, Bandar Lengeh, Iran High School Diploma of Physics and mathematics	Spring 2014

ACADEMIC & TEACHING EXPERIENCE

Kashmar Higher Education Institute, Kashmar, Iran Teacher Assistant	2017–2020
Assisted in <i>Introduction to Programming (C)</i>, <i>Advanced Programming (C++)</i> - Dr. Yousef Zadeh	
Head of Education, <i>Scientific Association</i> , Kashmar Higher Education Institute	2017–2020
Research Projects: Evolutionary Algorithms – Kashmar Higher Education Institute	2019
Implemented and presented optimization techniques based on evolutionary computation	
Research Projects: Genetic Algorithm – Kashmar Higher Education Institute	2019
Explored GA fundamentals and applications; presented findings at university	

SELECTED UNDERGRADUATE COURSE SCORES (SCALE: 20-POINT)

- Data Structures & Algorithms – 20.00
- Advanced Programming (C++) – 20.00
- Introduction to Programming (C) – 18.50
- Topics in Computer Science – 20.00
- Digital Logic Design – 20.00
- Linear Optimization – 15.00
- Data Mining – 20.00
- Scientific Computing Tools – 17.00
- Technical English for CS – 20.00
- Database Systems – 17.00
- Computer Systems – 15.50

PROFESSIONAL EXPERIENCE

Legend Company, Kashmar, Iran

2018–2020

Game Developer

- Worked in a startup game development team to build Uranus, a 2D educational platform game targeting elementary-school students.
- Created gameplay systems inspired by classic platformers like Super Mario, including movement physics, obstacles, collectibles, and progression mechanics.
- Integrated quiz-based learning modules into the levels to reinforce curriculum concepts through interactive challenges.
- Designed UI flow, game states, and event triggers to connect gameplay with educational quiz outcomes.
- Contributed to playtesting, balancing difficulty, and improving accessibility for younger age groups.

Team Size: 7

PROJECTS

Plato Mobile Game Automation System

2025

- Built an automation framework for mobile game interactions using Appium and Android emulators (LDPlayer/Nox), supporting 50+ game types with modular strategy-based architecture
- Implemented multi-instance parallel processing to run automation across multiple emulator instances simultaneously, improving throughput
- Designed extensible game strategy pattern with factory-based registration, enabling easy addition of new game types without modifying core logic
- Developed resource management system with automatic Appium server lifecycle management, health monitoring, and graceful error recovery with exponential backoff retry mechanisms
- Created thread-safe configuration management system with GUI-based configuration tool and JSON-based settings for flexible deployment
- Integrated image processing using OpenCV and PIL for UI element detection and game state recognition
- Built data management utilities for account creation, progress tracking, and automated daily task execution with scheduling capabilities

Tech: Python, Appium, Selenium

Code: Private repo (available on request) | **Team size:** 3

Telegram Automation Bot for Hamster Kombat & Web3 Clicker Games

2024

- Developed an automation bot for Hamster, as well as for TapSwap, Notcoin, Gemz, Dotcoin, and W-coin
- Implemented in Python, using Telegram Bot API + Web3 / TON blockchain integrations
- Automated repetitive “tap-to-earn” actions and task completion
- Handled multi-account management, concurrency, and error-handling to ensure bot stability over long-running sessions
- Built monitoring and logging so that all bot actions (earnings, tasks, wallet balances) were recorded, enabling performance tracking and debugging

Code: Private repo (available on request) | **Team size:** 3

Mobile & Desktop Application Developer, Lyndakade

2024

- Delivered a production-ready Flutter app spanning catalog, course detail, lists, and user profile flows
- Built a custom video player with background downloads and resume support
- Launched end-to-end payments (Zarinpal/Zibal) with cart, orders, and receipts
- Shipped search, notifications, and in-app chat (Goftino)
- Implemented QR course activation and robust empty/loading states for a smoother first-run experience
- Organized code into reusable widgets, screens, controllers, and services to speed feature delivery and maintenance
- Flutter (Dart), REST/JSON APIs, custom video player + background downloads, local storage/caching, push notifications, QR scanning, payment gateways (Zarinpal, Zibal), in-app chat (Goftino)

Code: Private repo (available on request) | **Team size:** 3

- Built a full-stack e-learning app using Laravel + Inertia.js Vue 3
- Implemented course management, chapters, videos, comments, and user progress tracking
- Implemented Vue 3 SSR via Inertia to improve SEO and initial render; managed the Node SSR service and deployment scripts
- Integrated fast, faceted search using Meilisearch + Laravel Scout; added type-ahead and relevance tuning
- Added authentication and access control with Laravel Fortify/Sanctum; implemented notifications and email flows
- Designed payment flows (multi-gateway), invoices, wallet/credit, and failure handling with idempotent verification
- Wrote feature/unit tests and optimized DB queries with eager loading and indexes; improved page TTFB and Core Web Vitals
- Automated deploys with bash/Windows scripts, system services for SSR, and environment-specific configs

Tech: Laravel, PHP, MySQL, Inertia.js, Vue 3, SSR, Tailwind, Meilisearch, Laravel Scout, Sanctum/Fortify, PHPUnit, NPM/Vite

Code: Private repo (available on request) | **Team size:** 3

Game Development for Educational Research, Kashmar Higher Education Institute

2017–2018

- Developed 3D car racing game for desktop platforms using Unity.
- Worked as a Game Developer on a research-led game under Ms. Dolkhani, contributing to an IEEE paper (ID 8712031).
- Architected game modules (levels, UI, logic) and implemented them using C#.
- Participated in design discussions to align game features with the research hypotheses and educational objectives.
- Conducted playtesting sessions, collected data, and refined gameplay based on user feedback and research requirements.

Team size: 4

Oxford Learning - Android Vocabulary App (Course Project), Kashmar Higher Education Institute

Fall 2017

- Built an Android application to browse and study IT vocabulary. Integrated a collaboratively compiled lexicon based on the Oxford English for Information Technology textbook, collected by the entire class under instructor supervision; implemented app structure, resource packaging, and data integration.

Tech: Android (Java), Android Studio

Code: [Github link](#)

PUBLICATIONS & RESEARCH PROJECTS

- Contributed to the development of a driving-based simulation game used in an IEEE-published study (Paper ID: 8712031), supervised by Ms. Dolkhani
- Analyzed potential improvements to HEFT-based load-balancing algorithm (referencing IEEE 7881902), supervised by Ms. Dolkhani

SKILLS

Programming: Python (advanced); PHP/Laravel, Java, JavaScript/Vue 3, Dart/Flutter

Data & ML: pandas, NumPy, scikit-learn, PyTorch (intro), TensorFlow/Keras (intro), EDA, data wrangling

Databases: MySQL/MariaDB (advanced)

Automation: Python scripting for web/desktop workflows

Version Control & Cloud: GitHub, GitLab; Google Cloud (basic)

Operating Systems: Linux, Windows

Microsoft Office Suite: Word, Excel, PowerPoint

COMPETITIONS

ACM Programming Competitions

- Sharif University, Tehran (2018)
- Ferdowsi University, Mashhad (2018)

LANGUAGES

Persian: Native

English: B2 (IELTS preparation in progress)

Spanish: A2 (on Duolingo, [View Profile](#))

PRESENTATIONS

Thesis Presentation: Lyndakade

- Kashmar Higher Education Institute
- **Year:** 2020
- **Evolutionary Algorithms**
 - Presented at Kashmar Higher Education Institute
 - **Year:** 2019
- **Genetic Algorithm**
 - Presented at Kashmar Higher Education Institute
 - **Year:** 2019

SPORTS

Karate-do: 2nd Dan Black Belt, April 2005 – 2016

Self-Defense: September 2019 - April 2020

Volleyball: February 2025 - Present

REFERENCES

Dr Atefeh Khazaei

Instructor (Database, Data mining) – role-model

Atefeh.Khazaei@port.ac.uk

Dr. Saeideh Yousefzadeh

Instructor (Introduction to Programming)

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Elham Dolkhani

Instructor – project leader – co-author

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Mohammad Mousavi Nezhad

Founder of Legend Startup – Head of Scientific Association at Kashmar Higher Education Institute

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